Bio ID: ___________ Per. _____

Protein Synthesis & Genetics Test Concept Review & Study Guide

- ☐ *Go over the sample questions*
- ☐ *Decide how confident you feel about each concept*
- ☐ *Pick an area to work on, and review it using the resources listed*
- ☐ *Repeat!*

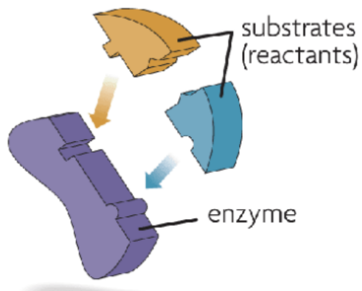
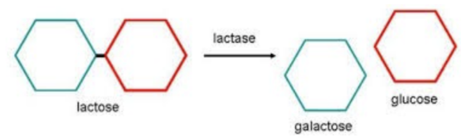
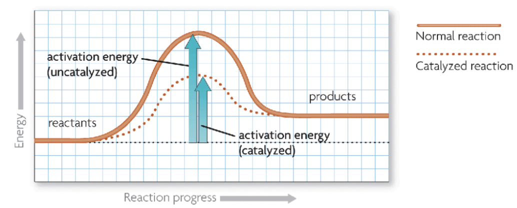
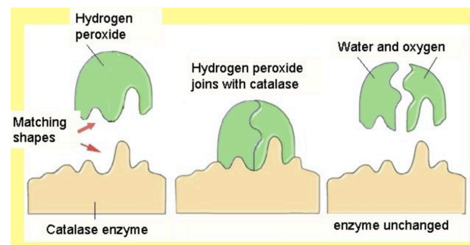
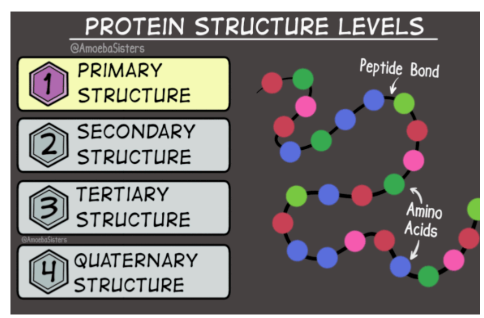
Concept	**Sample Question**	**Confidence Level?** 10= feeling great! 1= need to review this!	**Where to go to review** *** class sets of textbooks are available for review in the classroom during downtime, tutorial, and any dedicated studying time.*
Enzymes	• How do enzymes affect reactions, and how is the enzyme itself changed after affecting a reaction? • Describe how environmental factors affect enzymes.		• 1.2 Enzymes FIN • 1.2 Exploring Enzymes Lab • 2.1 Enzyme Review Sheet • Warm-up #27
Protein Synthesis	• Describe how information from DNA is used to form proteins. • What is the monomer of proteins? • Describe how changes to the DNA sequence can cause changes to a protein's function. • What molecules are read and what molecules are built during transcription? • During translation? • How does a ribosome know where to begin translating? • What are the differences between DNA and RNA? • <u>Use</u> a code wheel to translate an amino acid sequence. *Note: you do NOT need to memorize the code wheel!!*		• Warm-up #28 • 2.2 How Proteins are Made FIN • 3.1 Protein Synthesis Activity
DNA Structure	• What is the shape of a DNA molecule? • What is the monomer of DNA? • What are the 3 parts of this monomer? • Draw a labeled diagram of a short piece of a DNA molecule. • What are the names of the 4 nitrogenous bases? • Describe the base pairing rule.		• 3.1 DNA Structure Intro Notes • DNA Structure PPT Slides (on Schoology) • Warm-up #28

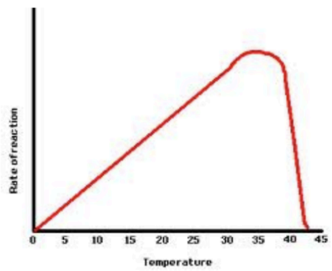
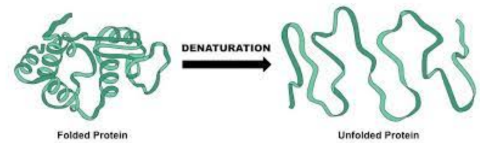
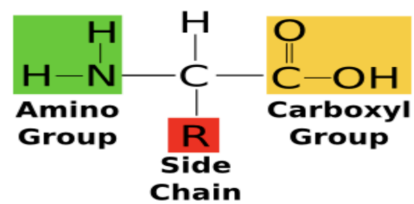
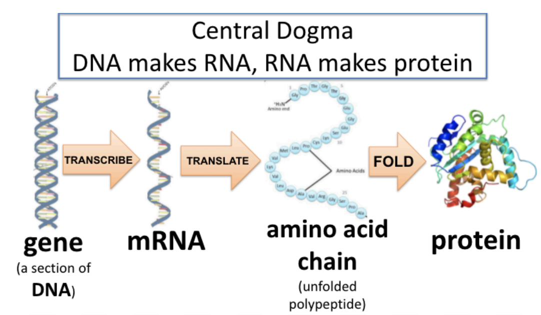
	• What type of bond connects nitrogenous bases?		
Mutations	• What is the difference between a point mutation and a frameshift? • How might each affect the amino acid sequence of a protein? • Describe several types of chromosomal alterations. • Explain how some mutations can be an advantage in one environment but a disadvantage in a different environment. • What is a mutagen? Give several examples.		• 3.2 Mutations FIN • 3.2 Mutations PPT Slides (on Schoology) • 3.1 Protein Synthesis Activity
Genetics Terms	• What is a gene made of? • What do we call different versions of the same gene? • What is the difference between a dominant and recessive allele? How did Mendel first see this? • What does it mean for a person to be heterozygous, homozygous dominant, or homozygous recessive for a trait? • What are the P, F_1, and F_2 generations?		• 3.3 Intro to Genetics FIN • 3.3 Intro to Genetics PPT Slides (on Schoology) • Textbook pp. 390-394
Genetics Problem Solving	• Solve problems similar to the problems on SpongeBob Genetics and the practice problems in class. • Label alleles, show crosses, complete Punnett squares, and report genotypic and phenotypic ratios from your Punnett squares.		• 3.3 Intro to Genetics FIN • Intro to Genetics PPT Slides (on Schoology) • 4.1 Intro to Punnett Squares • 4.1 SpongeBob Genetics • Warm-Up #30 & 31 • Monohybrid Crosses • Punnett Packet: #1-3
Other Dominance Patterns	• Given a word problem, identify a codominance or incomplete dominance inheritance pattern. • Explain the difference between codominance and incomplete dominance. • Label alleles for these types of problems (2 capital letters!)		• 4.2 Other Dominance Patterns FIN • 4.2 Other Dominance Patterns PPT Slides (on Schoology) • Punnett Packet: #4-8 • Warm-up #32 • Textbook p. 398

	● Solve these types of problems and report genotypic and phenotypic ratios.		
Multiple Alleles	● Explain the difference between a problem dealing with multiple alleles and any other problem we've worked with so far. ● Even if there are more than 2 possible alleles, how many alleles will each person inherit for every gene? ● Solve problems involving blood type inheritance: be familiar with the alleles and what genotypes go with what phenotypes. - for example: If someone has Type B blood, what possible genotypes could they have? Cross someone with Type B blood with someone who has Type AB blood. What phenotypes are possible for their children? ● Explain the connection between blood types and the antigens on blood cells. *Note: understanding the blood clotting relationships involved in blood transfusions is not required*.		● 5.1 Multiple Alleles & Polygenic Traits FIN ● 5.1 Multiple Alleles & Polygenic Traits slides (on Schoology) ● Warm-up #33 ● Punnett Packet #9-10 ● 5.2 Blood Typing Babies
Polygenic Traits	● Explain the difference between polygenic traits and other traits we've worked with. ● What are several examples of polygenic traits in humans?		● 5.1 Multiple Alleles & Polygenic Traits FIN ● 5.1 Multiple Alleles & Polygenic Traits slides (on Schoology)
Pedigrees	● Use a pedigree to show relationships in a family. ● Analyze a pedigree to determine which individuals are affected by a trait and which are carriers. ● Identify a sex-linked trait based on a pedigree showing who is affected.		● 5.2 Making Pedigrees (on back of Blood Type Babies) ● 5.2 Sex-linked Traits FIN ● 5.2 Sex-linked Traits slides (on Schoology) ● Punnett Packet #13
Sex-linked traits	● What makes a gene sex-linked? ● What is the difference between males and females in inheriting a		● 5.2 Sex-linked Traits FIN ● 5.3 Royal Disease

	recessive x-linked trait? Explain *why* these traits are more common in males than in females. • Solve problems involving sex-linked (recessive, on the X chromosome) traits.		• 5.2 Sex-linked Traits slides (on Schoology) • Punnett Packet #11-12 • Warm-up #34

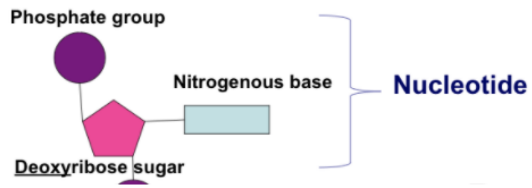
Proteins Synthesis: Vocabulary

Term	Definition	Example	Notes
Enzymes	Enzymes are proteins that act as **catalysts** for chemical reactions. (A catalyst speeds up chemical reactions.) Examples: lactase and catalase (*many enzymes end with “ase”)  **Substrates** are the reactant(s) in the chemical reaction. Examples: Hydrogen peroxide is a substrate, Lactose is also a substrate. The **active site** is the place where the substrate binds onto the surface of the enzyme.	  This figure shows how an enzyme acts as a catalyst to lower the amount of energy needed (activation energy) to start a chemical reaction. 	
Primary Structure	The primary structure of a protein is the amino acid sequence. This determines how the protein will fold and what shape it will be. Amino acids are held together by peptide bonds.		

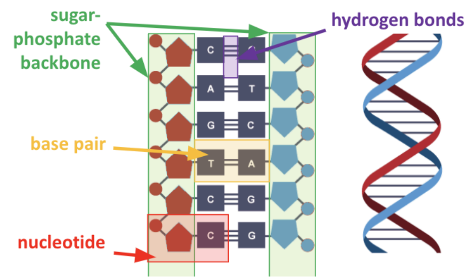
Denature	To denature an enzyme is to change the shape of an enzyme, due to **pH or temperature**. Changing the shape also changes the function. The graph below shows the rate of a reaction with a catalyst (an enzyme). The rate increases until the enzyme is heated too much. 	 The shape of an enzyme is important to its function. Environmental changes, like temperature or pH, can cause an enzyme to unfold and change shape. If an enzyme changes shape it may not be able to bind to the substrate.	
Protein	A protein is a folded chain of amino acids. • amino acid = monomer • protein = polymer The R group (or side chain) is different for each of the 20 different amino acids. The sequence of the side chains and their chemical properties determines how the primary structure bends, folds and coils.		
Protein Synthesis (how proteins are made)			

<table>
<tr>
<td>DNA
What is it?</td>
<td>

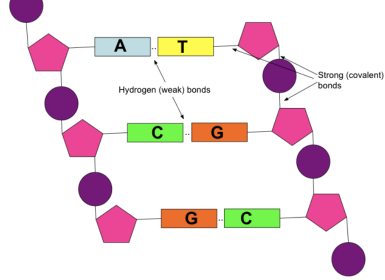
- Double-stranded
- Bases are A, C, G, T
- Contains all genetic information
- Has sugar deoxyribose
- Much longer than RNA (contains lots of genes)
- DNA cannot leave the nucleus
- Coiled into pieces called **chromosomes**
- Coding sections of DNA are called genes



</td>
<td>
What is the structure of DNA?



DNA is double stranded



</td>
<td>* note the types of bonds- weak hydrogen bonds between nitrogenous bases and strong covalent bonds between the phosphate groups and sugars</td>
</tr>
<tr>
<td>RNA
What is it?</td>
<td>

- Single-stranded
- Bases are A, C, G, U bases (has uracil instead of thymine)
- mRNA copies down genetic information from DNA and delivers it to the ribosome, so it can make proteins
- Has sugar ribose
- Much shorter than DNA (only contains information for making one protein)
- RNA can leave the nucleus

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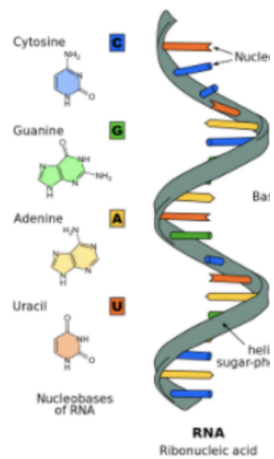

</td>
<td></td>
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</table>

Transcription

DNA to mRNA

(means to write down what is said)

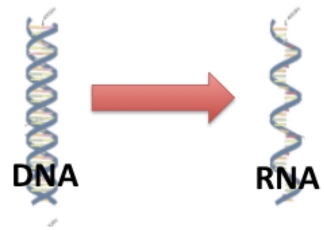
*occurs in the nucleus

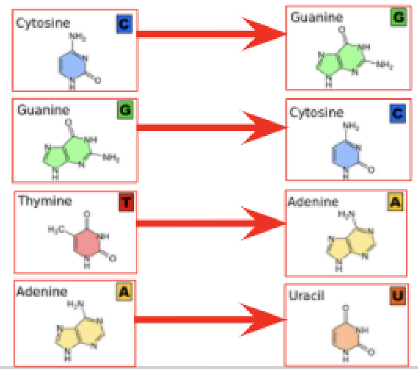
Transcription occurs in the nucleus where mRNA makes a copy of a gene (small section of DNA).

Only certain bases can pair together (called the “**Base Pairing Rule**”). “Straights” go together A and T (are drawn with straight lines) and “curves” go together C and G (drawn with curvy lines)

Base Pairing Rule

Template (starting) DNA:	T A C A A C G G T A C T
Daughter/coding DNA:	A T G T T G C C A T G A
mRNA:	AU G U U G C C A U G A





Translation

mRNA to protein

(convert from one language to another)
*occurs in the ribosome

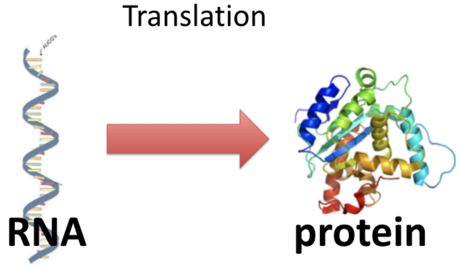
Amino acids are sequenced according to codons on mRNA.

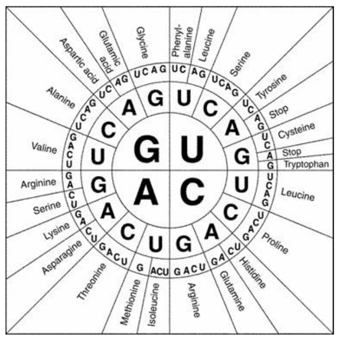
Codon: 3 bases that “code” for an amino acid.

Start codon: AUG (the amino acid methionine). A start codon is needed to start making a protein.

Stop codon: UAA, UAG, or UGA. This codon tells the ribosome to stop making the protein.

Translation





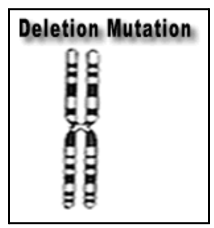
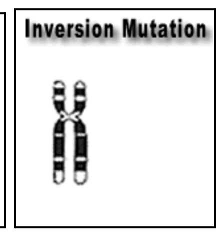
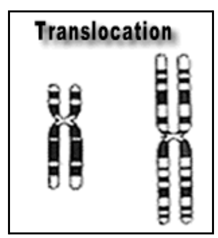
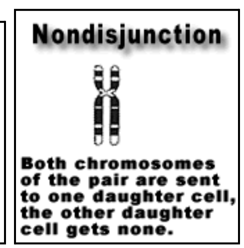
Second letter

First letter	U	C	A	G	Third letter
U	UUU, UUC Phe UUA, UUG Leu	UCU, UCC, UCA, UCG Ser	UAU, UAC Tyr UAA Stop UAG Stop	UGU, UGC Cys UGA Stop UGG Trp	U C A G
C	CUU, CUC, CUA, CUG Leu	CCU, CCC, CCA, CCG Pro	CAU, CAC His CAA, CAG Gln	CGU, CGC, CGA, CGG Arg	U C A G
A	AUU, AUC, AUA Ile AUG Met	ACU, ACC, ACA, ACG Thr	AAU, AAC Asn AAA, AAG Lys	AGU, AGC Ser AGA, AGG Arg	U C A G
G	GUU, GUC, GUA, GUG Val	GCU, GCC, GCA, GCG Ala	GAU, GAC Asp GAA, GAG Glu	GGU, GGC, GGA, GGG Gly	U C A G

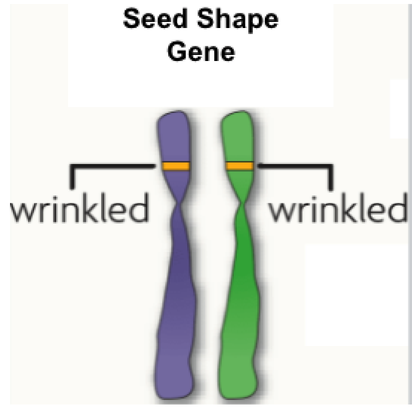
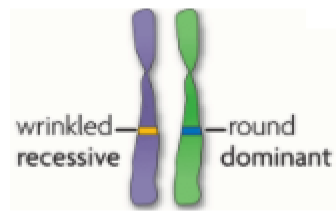
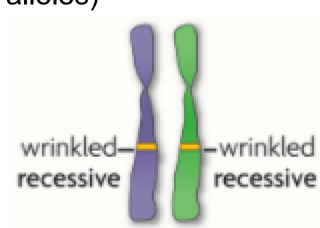
Use the key to translate this mRNA sequence: (remember not to start until you find the start codon AUG)

UAUGCCGAUUAACGCGUAGA

Mutations: Vocabulary

Mutations	Mutations happen when the sequence of DNA changes. They can be fatal, negative, benign (have no effect) or beneficial. **Somatic** mutation= in the body cell- not passed to the next generation **Gametic** mutation = occurs in the gametes, can be passed on the the next generation		
Errors in Replication	**Point mutation**- a substitution (one base pair is changed)- the other codons are unaffected **Frameshift mutation**- a single base pair is lost or added- all codons "downstream" are affected	**Point mutation** CAT ATE THE RAT **to** *CAT APE THE RAT* *one base pair changed **Frameshift mutation** CAT ATE THE RAT **to** *CAT APT ETH ERA T* *one base pair added	
Chromosomal Alterations	**Deletion**- part of the chromosome is left out **Inversion**- a piece of the chromosome breaks off and reattaches in reverse order **Translocation**- a broken piece attaches to a non-homologous chromosome **Duplication**- a piece of a chromosome is repeated **Non-disjunction**- a pair of chromosomes fails to separate during cell division (can result in a zygote with an extra chromosome)	   	
Mutagens	A mutagen is an outside agent that causes mutations. For example: UV rays, cigarette smoke, radiation, some chemicals and viruses.		

Intro to Genetics: Vocabulary

Term	Definition	Example	Notes
Gene	A **gene** is a piece of DNA that has the instructions to make a protein. **Genes** on homologous chromosomes are located in the same position. * This locus (or specific position) helps scientists study genes. The same **gene** can have many versions.		
Allele	An **allele** is any version of a gene that occurs at the same locus (position on a chromosome). **Dominant allele**: the allele that gets <u>expressed</u> when an individual has two *different* alleles. **Recessive allele**: the allele that gets <u>hidden</u> when an individual has two different alleles. *alleles are represented using letters (use capital and lowercase versions of the same letter)	**Heterozygous**: 2 different alleles (the example below has one wrinkled allele and one round allele)  The round allele is **dominant**. Dominant alleles are represented using capital letters **Homozygous:** 2 of the same allele (the example below has 2 wrinkled alleles)  The wrinkled allele is **recessive**.	

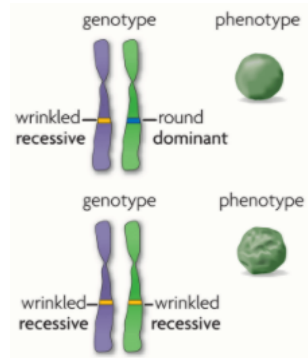
Genotype (Genes)

Genotype is an organism's genetic makeup (genes/alleles).

Examples: Bb or "heterozygous"

Phenotype (Features)

Phenotype is an expression of an organism's DNA; often it's appearance. Examples: blue eyes or black fur



R = allele for Pioneer Pride
r = allele for no Pioneer Pride

Genotype	Phenotype
RR	Pioneer Pride!
Rr	Pioneer Pride!
rr	No Pride ☹

Practice writing the genotype for the pea plant example:

Genotype	Phenotype
	round
	round
	wrinkled

*Remember, dominant alleles are capital and recessive are lowercase letters.

Punnett Square

Punnett squares are tables that show all the possible combinations of gametes.

Punnett squares are used to determine the probability of an offspring having a particular genotype.

*Make sure your teacher's can see that your letters are lowercase or capital.

mother

father

	B	b
B	BB	Bb
B	BB	Bb

Cross heterozygous Mom and homozygous dominant Dad.

Bb x BB

*B dominant for black fur, b recessive for brown fur.

Genotypic ratios of offspring:

- 2BB **:** 2Bb: 0bb
- 50% BB **:** 50% Bb: 0% bb
- ½ BB **:** ½ Bb: 0bb

Phenotype: 100% black fur, 0% brown

Mendel's Peas: tall plants are dominant (T) and short plants are recessive (t)

Use the Punnett Square to show a cross between two <u>heterozygous</u> pea plants.

<table>
<tr>
<td>Incomplete Dominance

(both alleles are dominant, both alleles are <u>capital</u> letters)</td>
<td>With incomplete dominance, a cross between organisms with two different phenotypes (like red and white flowers) produces an offspring with a third phenotype (pink flowers) that is a <u>blending</u> of parental traits.

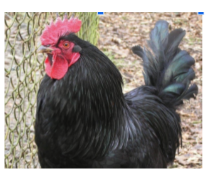
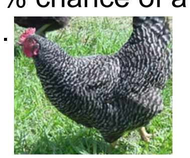
Both alleles show dominance, <u>both are assigned capital letters</u>.

What are all of the possible genotypes and phenotypes?
• RR = red
• WW = white
• RW = pink

Note: there are NO recessive alleles, so there are NO lower case letters used!</td>
<td>Cross a red and white flower (RR x WW)
 x 
Red four o'clock flowers White four o'clock flowers
<table><tr><td></td><td>R</td><td>R</td></tr><tr><td>W</td><td>RW</td><td>RW</td></tr><tr><td>W</td><td>RW</td><td>RW</td></tr></table>
Genotype ratio:
0% RR: 100% RW: 0% WW

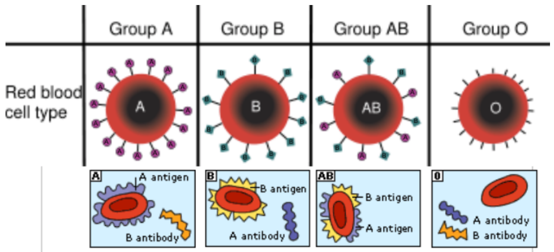
Phenotype ratio:
0% red: 100% pink: 0% white</td>
<td></td>
</tr>
<tr>
<td>Codominance

(both alleles are dominant, both alleles are <u>capital</u> letters)</td>
<td>With codominance, a cross between organisms with two different phenotypes produces offspring with a <u>third</u> phenotype in which both of the parental phenotypes appear <u>together, side by side</u>.

Example: A black chicken (BB) and white (WW) chicken
 x 
= would have a 100% chance of a checkered offspring.
</td>
<td>Cross a white chicken and checkered chicken (WW x BW).
 x 
<table><tr><td></td><td>B</td><td>W</td></tr><tr><td>W</td><td>BW</td><td>WW</td></tr><tr><td>W</td><td>BW</td><td>WW</td></tr></table>Genotype ratio:
0% BB: 50% BW: 50% WW

Phenotype ratio:
0% black: 50% checkered: 50% white</td>
<td></td>
</tr>
</table>

<table>
<tr>
<td>Multiple Alleles</td>
<td>Genes can multiple (have more than 2) alleles.

Blood Type is an example of multiple alleles.

</td>
<td>Possible Blood Types: A, B, O, AB

Allele notation: I^A I^B i
– I^A allele is dominant to i
– I^B allele is dominant to i
– I^A and I^B alleles are codominant

What are the possible genotypes for each of the phenotypes?

Type A: I^AI^A, I^Ai
Type B: I^BI^B, I^Bi
Type AB: I^AI^B
Type O: ii</td>
<td></td>
</tr>
<tr>
<td>Polygenic Traits</td>
<td>Polygenic traits are traits that are controlled by more than one gene.

Examples: eye color, human height, skin color, fingerprint pattern, hair color.

* offspring receive more than 2 alleles from one parent and more than 2 alleles from another parent that determine a specific trait. Example: BB bb x bb Gb</td>
<td>Example: Two different genes control eye color inheritance. *Remember capital letters show dominance.
<table>
<tr><th>Genotype</th><th>Phenotype</th></tr>
<tr><td>BB bb</td><td>Brown</td></tr>
<tr><td>BB Gb</td><td>Brown</td></tr>
<tr><td>BB GG</td><td>Brown</td></tr>
<tr><td>Bb bb</td><td>Brown</td></tr>
<tr><td>Bb Gb</td><td>Brown</td></tr>
<tr><td>Bb GG</td><td>Brown</td></tr>
<tr><td>bb GG</td><td>Green</td></tr>
<tr><td>bb Gb</td><td>Green</td></tr>
<tr><td>bb bb</td><td>Blue</td></tr>
</table></td>
<td></td>
</tr>
<tr>
<td>Sex Linked Traits

(Y and X chromosomes)</td>
<td>Genes on sex chromosomes are called sex-linked genes.

*Males have an XY genotype (all of a males sex-linked genes are expressed). Males cannot be carriers of a gene!

**Females have an XX genotype.</td>
<td>We show sex-linked genes on the chromosome they are found on- either an X or a Y.

Remember this example of hemophilia:
X^HY x X^HX^h

(cross between a non-hemophilic male and a female who is a carrier)</td>
<td></td>
</tr>
</table>

Pedigrees	Pedigrees are used to map traits across many generations and in extended families. Genetic counselors use pedigrees to help their clients understand the probability of passing heritable traits to the next generation. * Squares represent males and circles represent females. ** Fully shaded boxes mean that person has the trait. Boxes that are half shaded mean that person is a carrier.